

One-Variable Inequalities: Catch the Sign Flip

Directions:

- Solve each inequality and show every step in the work space.
- Whenever you multiply or divide both sides by a **negative** number, the inequality symbol must **reverse**. Watch for it!
- Two problems show another student's work that contains a mistake. Find the mistake, then fix it.

1. Solve the inequality: $6x > 42$

Answer: _____

2. Solve the inequality: $-5x \leq 30$

Answer: _____

3. Solve the inequality $-4x < 18$.

(a) Solution: _____

(b) The *least integer* that is a solution: _____

4. Solve the inequality: $3x - 7 > 5$

Answer: _____

5. Solve the inequality: $-2x + 9 \leq 1$

Answer: _____

6. Priya solved an inequality like this:

$$-3x < 12 \Rightarrow \frac{-3x}{-3} < \frac{12}{-3} \Rightarrow x < -4$$

(a) What went wrong? Explain in one sentence.

(b) Solve the inequality correctly: _____

7. Deshaun solved an inequality like this:

$$-2x + 3 > 11 \quad \Rightarrow \quad -2x > 14 \quad \Rightarrow \quad x < -7$$

(a) Find the step with the error and say what went wrong.

(b) Solve the inequality correctly: _____

8. The temperature in a freezer experiment starts at 14°F and drops 3°F every hour, so after h hours the temperature is $14 - 3h$ degrees.

(a) Write and solve an inequality to find when the temperature will be *below* -7°F .

Solution: _____

(b) Your solving used division by a negative number. Explain why the inequality symbol had to reverse.